

IMPLICATIONS OF DIGITAL TECHNOLOGIES

Digital technologies have fundamentally changed how society shares and uses information, music, images, and films. While they provide powerful tools for development, they introduce significant societal challenges that force us to rethink:

- Ownership and Control
- Privacy and Identity
- Censorship
- Health, Safety, and Environmental Impacts

Data Protection and Rights

Legal Frameworks

Some countries enact specific laws to ensure that personal data stored by organizations remains protected.

- **Singapore (Personal Data Protection Act):** Requires organizations to obtain an individual's explicit consent to collect, use, or disclose personal data. Data must only be used in an appropriate manner for informed, specific purposes that a reasonable person would consider appropriate under the circumstances.
- **United Kingdom (Data Protection Act):** Restricts data use to strict data protection principles. Organizations must ensure personal data is:
 - Used fairly and lawfully.
 - Used for limited, specifically stated purposes.
 - Adequate, relevant, and not excessive.
 - Accurate
 - Kept no longer than necessary.
 - Handled according to individual data rights
 - Kept safe and secure.
 - Not transferred outside the European Economic Area (EEA) without adequate protection.

Individual Legal Rights

Under legislation like the UK Data Protection Act, individuals have the right to:

1. Access a copy of their stored personal data.
2. Object to data processing that causes damage or distress.

3. Prevent data processing used for direct marketing (e.g., newsletters, automated promotional emails).
4. Object to decisions made entirely by automated systems.
5. Have inaccurate personal data rectified, blocked, erased, or destroyed.
6. Claim financial compensation for damages caused by an organizational breach of the Act.

International Agreements

Because data is often stored across international border lines, agreements are crucial to ensure organizations act lawfully.

- **The Challenge:** Depending on the online services used, user data might be stored in countries with weaker or non-existent data protection laws, leaving individuals with no legal recourse if their rights are abused.
- **The Solution:** Agreements like the **EU-US Privacy Shield Framework** aim to protect data transferred across borders (e.g., between the US and the EU), which is vital for multinational technology companies with massive data operations.

Copyright Legislation

Concept of Copyright

Copyright is a legal power providing creators of original work with the sole authority to use and distribute it. If users download digital assets (software, books, music, images, or videos) without permission in a jurisdiction protected by copyright law, they face potential legal action from the copyright owner.

Distribution Licences

Owners can choose how they share their work:

- **Paid Licences:** Users pay for the right to use the media (e.g., buying a track or app from an online store), but this does not grant the right to distribute it to others.
- **Free Licences (Open Source / Creative Commons):** Creative Commons licences allow owners to distribute their work freely without payment, while still protecting their rights by explicitly specifying how other people can use, share, or alter the work. Free software licences lower the barrier to technology access, helping narrow the **digital divide** and the **information gap**.

Copyright Exemptions (Fair Use)

In many countries, certain conditions allow exceptions where copying materials without the explicit permission of the owner is lawful:

- Digital or physical copies made by libraries, educational establishments, museums, or archives.
- Record keeping of broadcasts intended specifically for archive purposes.
- Alternative formats developed directly for the benefit of people with a disability.
- Non-commercial research or private study.
- Material utilized for the purpose of a parody (exaggerated humorous imitation).

Digital Rights Management (DRM) and Geoblocking

To prevent illegal distribution and check whether a device is authorized to open a file, owners use copy protection and **Digital Rights Management (DRM)** systems (e.g., files downloaded on Apple iTunes). Additionally, organizations implement **geoblocking** to limit access to internet content based on the user's geographical location, preventing users in different legal frameworks from accessing copyrighted work.

Monitoring Individuals

Monitoring Movements

Governments and organizations can track individual locations using various mechanisms:

- **CCTV:** Closed-circuit television networks, often augmented with facial recognition software.
- **ANPR:** Automatic Number Plate Recognition cameras used by law enforcement.
- **Card Transactions & IDs:** Monitoring border crossings via passports, travel cards, identification cards, and bank transactions.
- **Network Tracing:** Identifying device footprints on networks via IP or MAC addresses.
- **GPS Technology:** Utilizing real-time GPS data shared by mobile applications or tracker devices clipped to clothing.

Benefits vs. Drawbacks of Movement Monitoring

- **Benefits:**
 - Locating lost individuals (such as young children),
 - organizing social gatherings via friend-finder apps,
 - identifying people on network ,
 - tracking down potential criminals at public events or airports,
 - verifying identities to lower financial crime rates.
- **Drawbacks:**
 - Severe compromises to personal privacy,
 - high financial expenses to establish and maintain systems,
 - heavy energy consumption impacting the environment,
 - causing citizens to feel distrusted.

Countermeasures (Anti-Surveillance)

To fight back against tracking, some modern fashion designs function as camouflage:

- **Anti-Surveillance Clothing:** Patterns featuring shapes that facial recognition software misinterprets as a face, overloading the system with data and hiding the true face of the individual.
- **Anti-Surveillance Eyewear:** Glasses specifically built to reflect the infra-red light used by CCTV cameras during low-light capture, blocking them from recording facial dimensions.

Monitoring Communications

Digital interactions—such as text messages, emails, social media posts, blog comments, and web history—can be systematically logged. Parents often use parental control software to check their children's activities. In certain jurisdictions, laws like the UK's *Investigatory Powers Act 2016* legally permit police and security agencies to access digital communications, sparking major civil liberties and human rights controversies regarding the right to privacy.

Sustainability Issues

The growing infrastructure of Information and Communication Technology (ICT) places immense strain on global ecology.

- **Sustainability:** Not using up natural resources until they run out; maintaining the balance of nature
- **Mitigation:** Reducing severity of a problem

Key Sustainability Challenges and Mitigations

Cause of Issue	Sustainability Problem	Mitigation Strategies
Power Requirements of Digital Devices	Electricity generated using non-renewable energy sources (coal, gas) depletes the planet's natural resources.	• Transitioning to renewable energy sources. • Manufacturing and using more energy-efficient hardware.
Cooling Systems in Data Centres	Massive server farms require high-power artificial cooling setups that consume non-renewable fuel.	• Building data centres next to rivers to harness natural hydroelectric power. • Constructing data centres in naturally cold climates (e.g., Northern Sweden) to eliminate the need for high-energy air conditioning units. • Using recycled canal water or seawater for cooling systems.

Use of Poisonous Substances	Digital components contain harmful chemicals like bromine, mercury, and chlorine. When improperly discarded as e-waste, these leak into soil and water systems, causing wildlife destruction and severe human health conditions (cancer, lung disease, central nervous system damage).	• Promoting public education and recycling initiatives. • Adhering to strict international directives (e.g., the EU's Waste Electrical and Electronic Equipment Directive). • Engineering components using harmless alternative materials.
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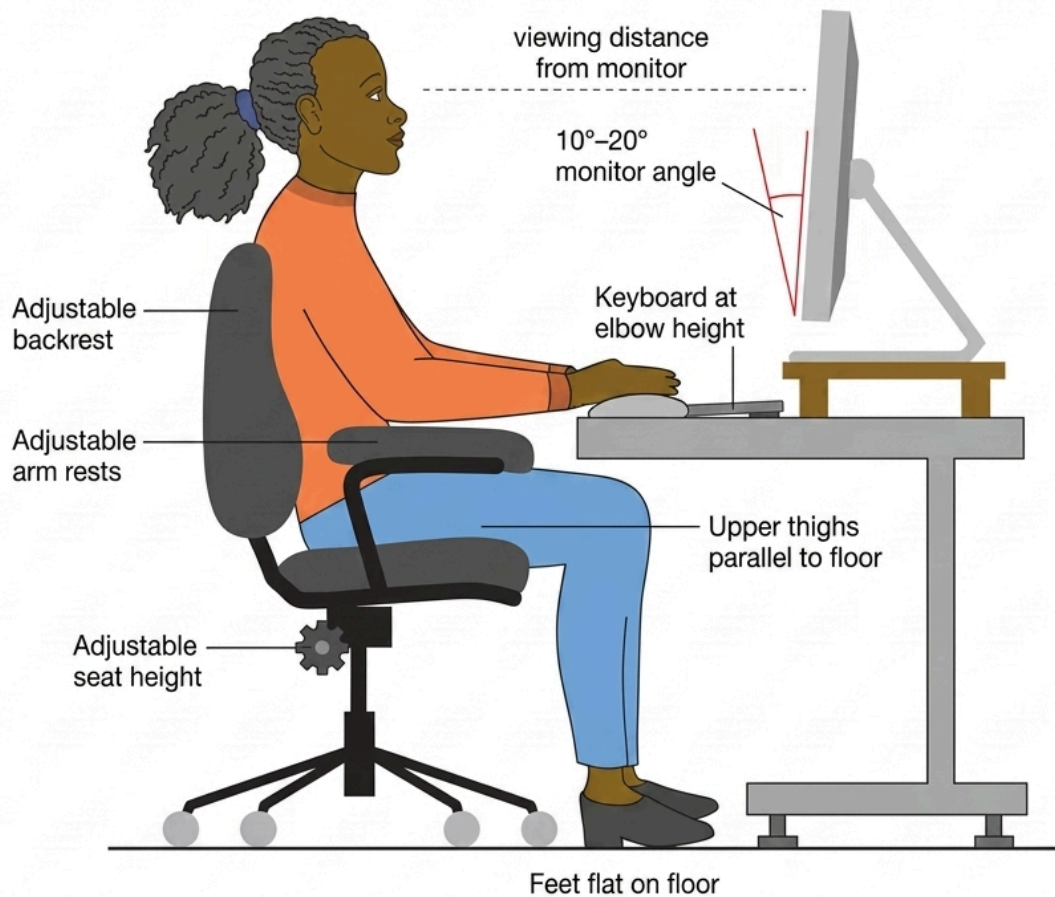
Health and Safety Issues

Prolonged or improper use of digital devices can lead to distinct occupational hazards.

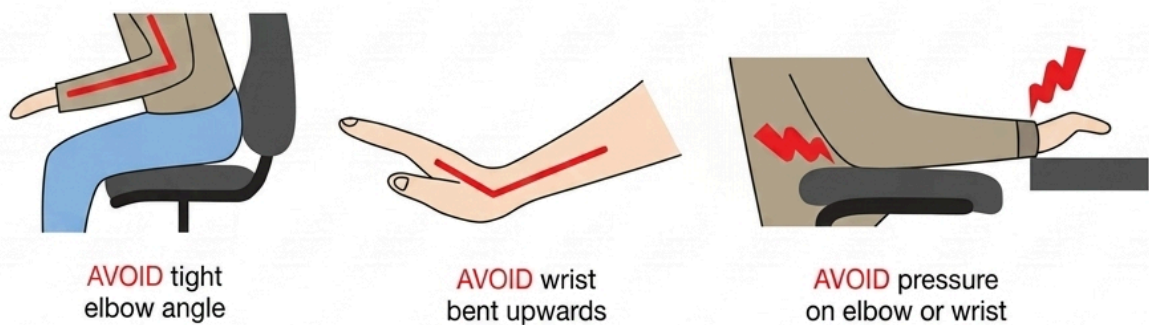
Risks, Causes, and Minimization Methods

- **Eye Dryness and Fatigue**
 - *Cause:* Looking at screens for extended periods without blinking. Human eyes are sensitive to short-wavelength blue light emitted by LEDs, which can disrupt sleep cycles.
 - *Minimization:* Take frequent breaks to look into the distance, ensure screens are kept at an appropriate viewing distance, use adequately sized monitors, apply blue light filters, and use glare-reducing lighting.
- **Repetitive Strain Injury (RSI) / Carpal Tunnel Syndrome**
 - *Cause:* Repeating the same physical actions continuously while using devices incorrectly or maintaining poor posture, compressing the wrist nerve.
 - *Minimization:* Use ergonomically designed devices and supportive peripherals, such as cushioned wrist pads.
- **Back and Neck Ache**
 - *Cause:* Slouching or maintaining an incorrect sitting posture over extended periods.
 - *Minimization:* Sit with a proper posture: upper thighs parallel to the floor, feet flat, keyboard positioned at elbow height, and an adjustable backrest.
- **Trip Hazards**
 - *Cause:* Trailing or loose power and data cables across walkways.
 - *Minimization:* Establish structured cable management by securing wires inside defensive plastic trunking channels.
- **Electric Shock**
 - *Cause:* Damaged external wire insulation or spilling liquids directly onto active electronic devices.
 - *Minimization:* Perform regular inspection of cables, immediately repair/replace worn wires, keep liquids away from workstations, and use Residual Current Devices (RCDs) to interrupt faulty circuits.
- **Fire Hazards**
 - *Cause:* Hardware overheating due to blocked ventilation or overloading electrical plug sockets with too many high-draw adapters.

- *Minimization*: Install dedicated active cooling devices, conduct scheduled maintenance, fit sockets with appropriate internal fuses, and keep fire extinguishers accessible nearby.
- **Physical Injury or Death**
 - *Cause*: Sudden lack of environmental awareness due to severe mobile distractions (e.g., texting while walking or driving).
 - *Minimization*: Strictly refrain from interacting with digital screens when interacting with roads or navigating immediate physical hazards.



▲ **Figure 9.7** Maintaining good posture when sitting at a computer



▲ **Figure 9.8** Incorrect posture can cause health problems

Artificial Intelligence (AI) and Risk

As artificial intelligence automation takes over daily human decisions (e.g., autopilot systems in airplanes or self-driving cars applying brakes, steering away from lane drift, and monitoring road speed limits), it introduces a critical ethical dilemma.

AI applications operate on programmatic rules rather than human common sense or moral judgment. This creates complex issues:

- **The Speed vs. Judgment Paradox:** While computers react significantly faster than human drivers—potentially reducing the 94% of accidents caused by human error—they cannot flexibly break a rule to avoid an unexpected hazard (e.g., safely exceeding a speed limit to escape danger).
 - **The Ethical Dilemma:** If an automated car must swerve to avoid a collision but accidentally strikes a pedestrian in the process, a machine cannot be held ethically or morally responsible for that action.
 - **The Attention Lapse:** When automated systems take long-term control, human operators experience degraded reaction times or may even fall asleep, leaving them unable to react during critical emergencies.
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End of chapter questions

- 1) Which one of these is used to reduce the risk to health and safety? (1)
 - a) Using a small monitor
 - b) Increasing the amount of blue light emitted by a screen
 - c) Using an RCD
 - d) Working in a room with low-level lighting
- 2) Explain why free software licences have helped to reduce the digital divide. (2)
- 3) Explain why users should be able to give their consent before an app shares their location. (2)
- 4) State three types of digital information that can be protected by copyright. (3)
- 5) State one method of monitoring an individual's movements and describe how it is used. (2)
- 6) Explain why owners of digital information may decide to use copyright. (2)
- 7) State one benefit to individuals of having their location monitored. (1)
- 8) State one benefit to individuals of having their communications monitored. (1)
- 9) State one drawback to individuals of monitoring their location and communications. (1)
- 10) State three ways in which organisations should act responsibly to make sure that they protect the data they hold about individuals. (3)
- 11) Explain why the large-scale use of online storage and applications can have a negative effect on the environment. (4)
- 12) Describe one way in which individuals can reduce the effect of their use of digital devices on the environment. (3)